Sri Lanka Institute of Information Technology



RideLanka FleetHire

Web-based Car Rental Management System

Lab Submission – 04

Activity Diagram & System Function Distribution

2026-Y2-S1-KU-36

Group Members and Core Ownership

No.	Member / Registration No.	Assigned Function
01	S M H C B D Rambukpitiya - IT25103736	User & customer management
02	I K S P Jayawardana - IT25103732	Vehicle management
03	W M K K G N A Weerasingha - IT25103737	Booking & reservation management
04	R M M D Bhagyamathi - IT25103733	Payment management
05	R M R D Rajapaksha - IT25103734	Feedback & support management
06	K A K Dananjaya - IT25103735	Fleet management

Software Engineering | SE2030

B.Sc. (Hons) in Information Technology

01 User & customer management Activity Diagram

UM04 - Suspend a Customer Account for Policy Violation

OWNERSHIP AND TRACEABILITY S M H C B D Rambukpitiya - IT25103736 | Core ownership: User & customer management | Diagram use case: UM04

Use Case Definition

Use case ID	UM04
Use case name	Suspend a Customer Account for Policy Violation
Goal in context	Prevent a customer who has violated platform policy from making new bookings, while preserving an accountable status history.
Scope	RideLanka FleetHire - User & Customer Management
Level	User goal
Summary	A System Administrator changes an existing customer account to SUSPENDED through a controlled, version-checked status transition.
Priority	High - trust and safety
Primary actor	System Administrator
Supporting actors	None
Stakeholders and interests	Administrator needs an accurate account record; Front-Desk and Reservations staff need suspended customers blocked from new bookings; auditors need a traceable reason and time.
Trigger	The System Administrator determines that a customer account must be restricted following a policy violation.
Preconditions	The Administrator is signed in and active; the customer account exists; its current version is displayed; a reason can be supplied; the account is not already SUSPENDED.
Success postcondition	The account is SUSPENDED, the reason and status history are retained, and the customer cannot create new bookings.
Failure guarantee	The account remains in its last valid state; no unrecorded or partially applied status change is left behind.
Related use cases	UM01 Register Customer Account; UM02 Sign In or Recover Account; UM03 Manage Profile and Permissions; BR01 Create Online Booking.
Implementation alignment	Aligned to the current Administrator command. Broader Customer Service escalation workflow is intentionally not presented as already implemented.

01A Main Success Scenario

UM04 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The Administrator opens the selected customer account record.
2	The system displays the current account status, version, and recent status history.
3	The Administrator chooses Suspend Account and enters the reason.
4	The system verifies that the actor is an active Administrator and that all required information is present.
5	The system confirms that the account still exists and that the displayed version is current.
6	The system confirms that the requested transition from the current state to SUSPENDED is valid.
7	The system changes the account status to SUSPENDED.
8	The system records the old status, new status, reason, actor, and time in the status and audit history.
9	The system shows the updated account and blocks the customer from creating any new booking.

01B Alternative and Exception Flows

UM04 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
3a	Reason is missing	The system identifies the required field and keeps the account unchanged.
4a	Actor is inactive or unauthorised	The system denies the action without revealing restricted account data.
5a	Account no longer exists	The system reports that the account cannot be found and makes no change.
5b	Another user changed the account	The system reports a version conflict and asks the Administrator to reload the latest record.
6a	Account is already SUSPENDED	The system rejects the invalid repeated transition and displays the current state.
8a	The change or audit record cannot be saved	The complete operation is cancelled and the prior account state remains authoritative.

Business Rules

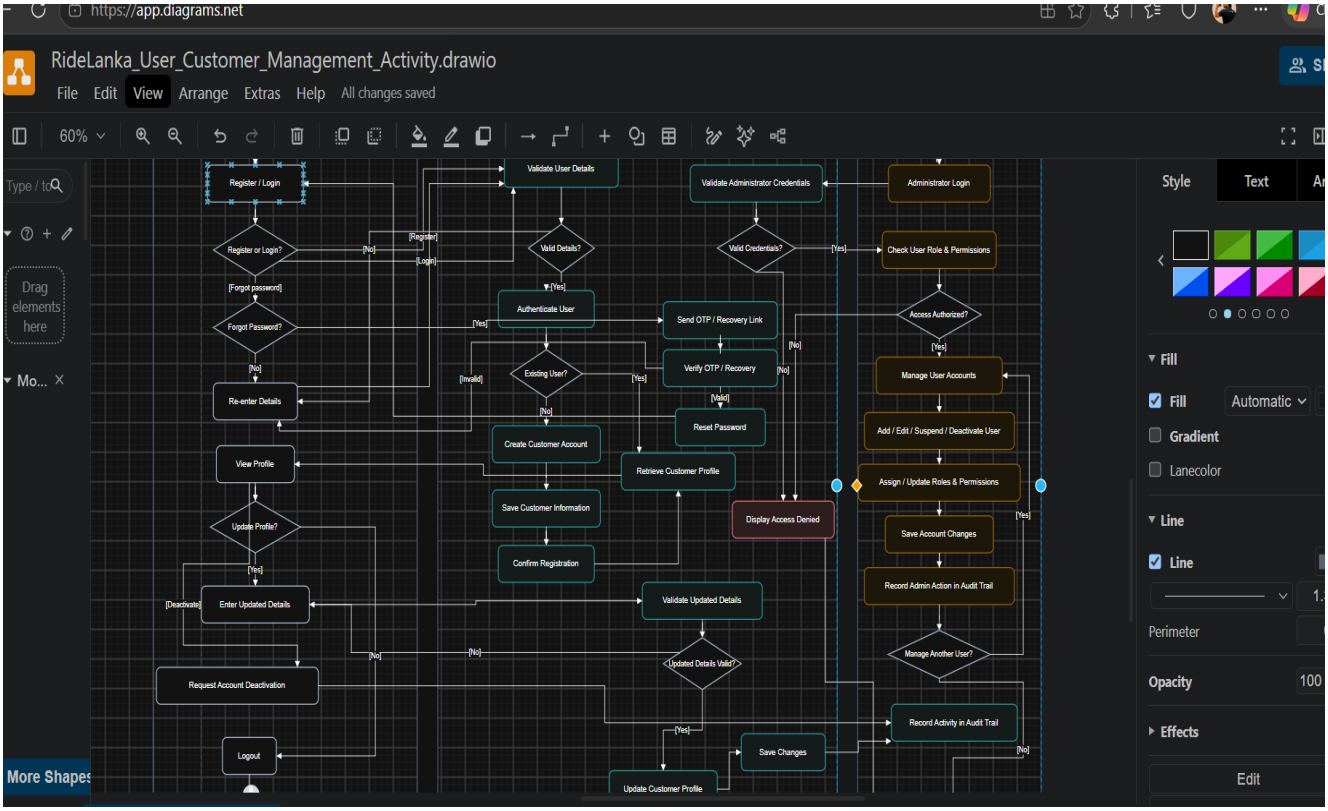
- Only an active System Administrator may perform this account-status command.
- A reason and the current record version are mandatory.
- A SUSPENDED customer cannot create, pay for, or confirm a new booking.
- The old state, new state, reason, actor, and time must remain traceable.
- Account history is retained; permanent deletion is not used for this status event.

Special Requirements and Open Decisions

Security: enforce role checks for every request. Reliability: the status and audit history must either both succeed or both remain unchanged. Usability: show the latest version-conflict message clearly. Open issues: guest checkout linkage, appeal workflow, and automated suspension notification remain future work.

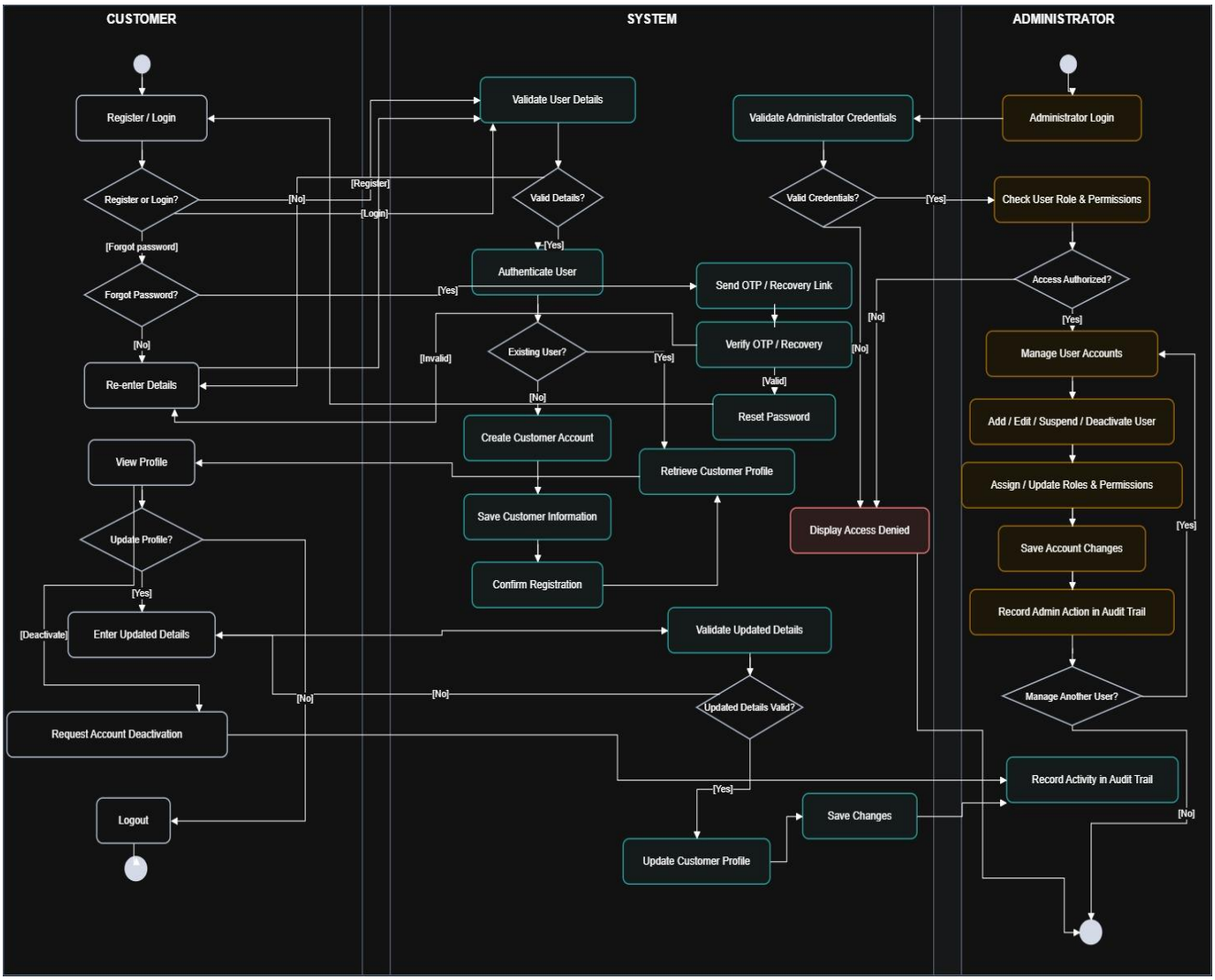
01C Draw.io Proof

EVIDENCE PURPOSE This space is reserved for a screenshot verifying that the activity diagram was authored and reviewed through Draw.io. Take a screenshot of your Draw.io canvas for this use case and insert it below.



01D UM04 Activity Diagram - Clean Export

Swimlanes: System Administrator and RideLanka FleetHire System.



02 Vehicle management Activity Diagram

VM03 - Mark a Vehicle Out of Service

OWNERSHIP AND TRACEABILITY I K S P Jayawardana - IT25103732 | Core ownership: Vehicle management | Diagram use case: VM03

Use Case Definition

Use case ID	VM03
Use case name	Mark a Vehicle Out of Service
Goal in context	Prevent an unsafe or unusable vehicle from receiving new bookings while preserving an accountable status history.
Scope	RideLanka FleetHire - Vehicle Management
Level	User goal
Summary	A Fleet Management staff member changes an existing vehicle to OUT_OF_SERVICE through a controlled, version-checked status transition.
Priority	High - safety and operational continuity
Primary actor	Fleet Management Staff
Supporting actors	None
Stakeholders and interests	Fleet staff need an accurate vehicle record; Reservations staff need unavailable vehicles excluded from search; auditors need a traceable reason and time.

Trigger	Fleet Management staff determine that a vehicle must be removed from active availability.
Preconditions	The staff member is signed in and active; the vehicle exists; its current version is displayed; a reason can be supplied; the vehicle is not already OUT_OF_SERVICE.
Success postcondition	The vehicle is OUT_OF_SERVICE, the reason and status history are retained, and future search and booking cannot use the vehicle.
Failure guarantee	The vehicle remains in its last valid state; no unrecorded or partially applied status change is left behind.
Related use cases	VM01 Register and Categorise Vehicle; VM02 Update Vehicle Details and Rates; VM04 View Vehicle Status History; BR02 Check Vehicle Availability.
Implementation alignment	Aligned to the current Fleet Management command. The broader incident and insurance-claim workflow is intentionally not presented as already implemented.

02A Main Success Scenario

VM03 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The Fleet staff member opens the selected vehicle record.
2	The system displays the current operational status, version, and recent status history.
3	The staff member chooses Mark Out of Service and enters the reason.
4	The system verifies that the actor is an active Fleet Management staff member and that all required information is present.
5	The system confirms that the vehicle still exists and that the displayed version is current.
6	The system confirms that the requested transition from the current state to OUT_OF_SERVICE is valid.
7	The system changes the vehicle status to OUT_OF_SERVICE.
8	The system records the old status, new status, reason, actor, and time in the status and audit history.
9	The system shows the updated vehicle and excludes it from any later availability or booking check.

02B Alternative and Exception Flows

VM03 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
3a	Reason is missing	The system identifies the required field and keeps the vehicle unchanged.
4a	Actor is inactive or unauthorised	The system denies the action without revealing restricted fleet data.
5a	Vehicle no longer exists	The system reports that the vehicle cannot be found and makes no change.
5b	Another user changed the vehicle	The system reports a version conflict and asks the staff member to reload the latest record.
6a	Vehicle is already OUT_OF_SERVICE	The system rejects the invalid repeated transition and displays the current state.
8a	The change or audit record cannot be saved	The complete operation is cancelled and the prior vehicle state remains authoritative.

Business Rules

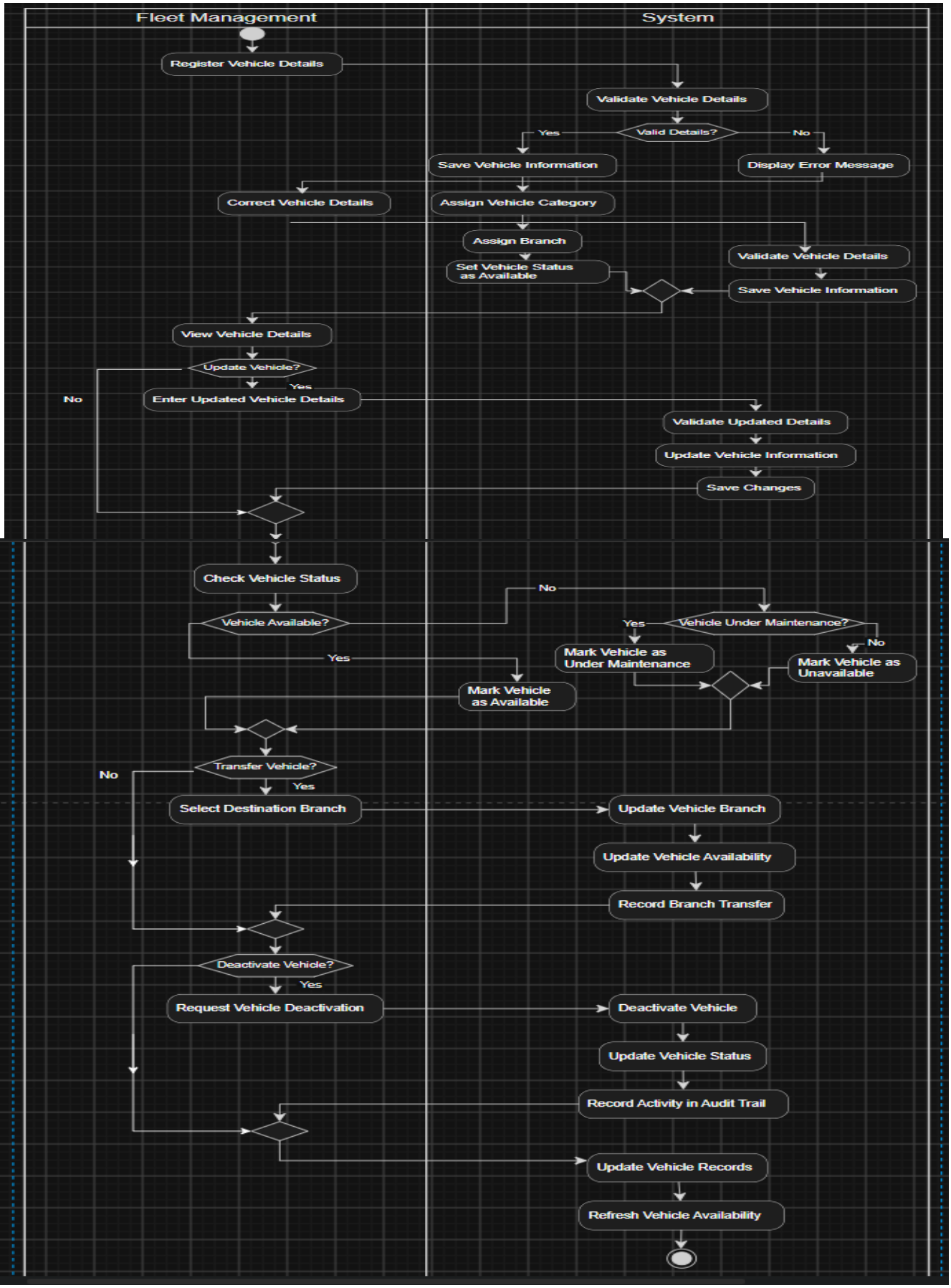
- Only active Fleet Management staff may perform this vehicle-status command.
- A reason and the current record version are mandatory.
- An OUT_OF_SERVICE vehicle cannot be booked or shown as available at any branch.
- The old state, new state, reason, actor, and time must remain traceable.
- Operational history is retained; permanent deletion is not used for this status event.

Special Requirements and Open Decisions

Security: enforce role checks for every request. Reliability: the status and audit history must either both succeed or both remain unchanged. Usability: show the latest version-conflict message clearly. Open issues: cross-branch transfer during out-of-service status, insurance verification linkage, and automated customer reassignment remain future work.

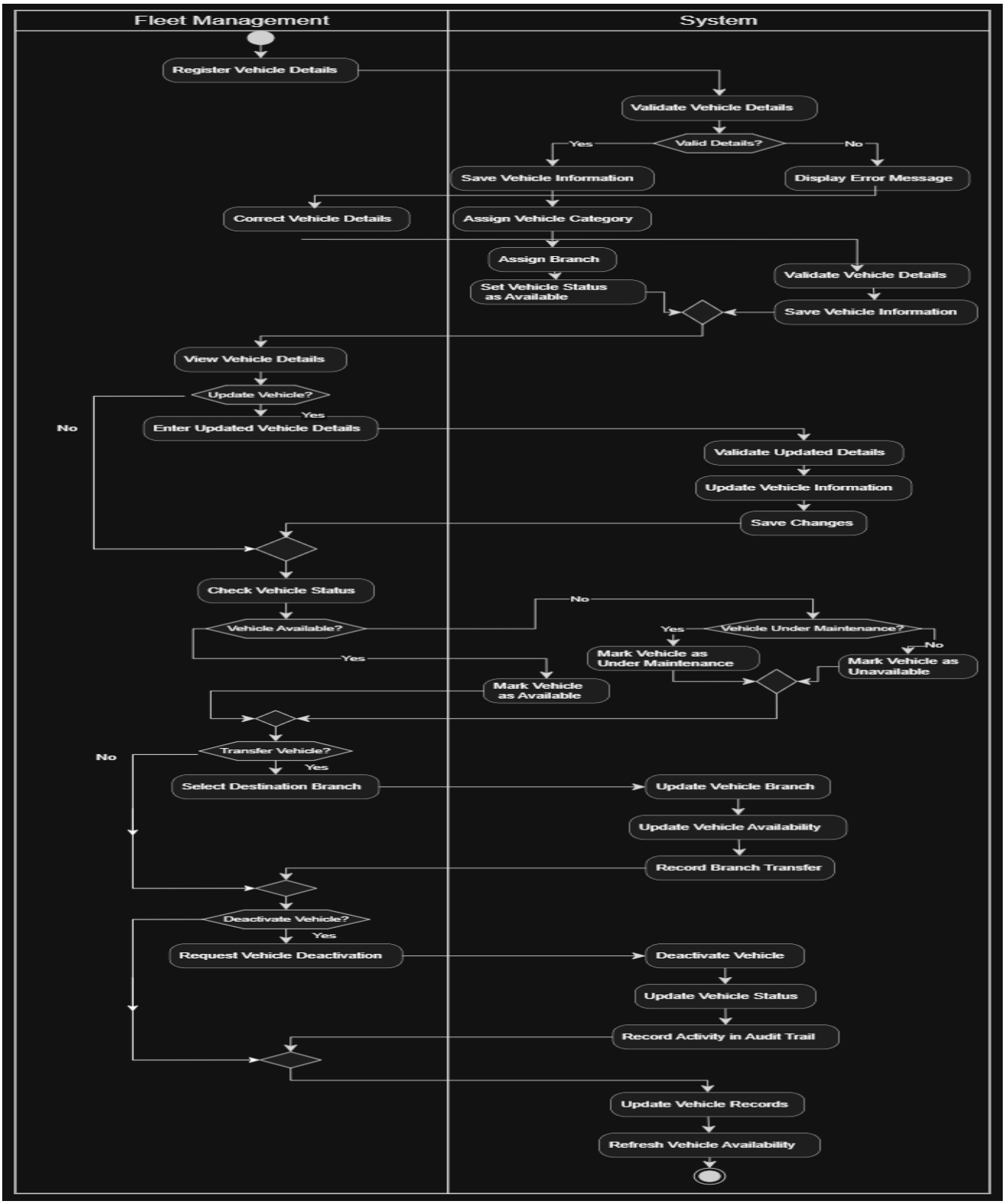
02C Draw.io Proof

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02D VM03 Activity Diagram - Clean Export

Swimlanes: Fleet Management Staff and RideLanka FleetHire System.



03 Booking & reservation management Activity Diagram

BR04 - Reserve a Vehicle and Confirm a Booking

OWNERSHIP AND TRACEABILITY W M K K G N A Weerasingha - IT25103737 | Core ownership: Booking & reservation management | Diagram use case: BR04

Use Case Definition

Use case ID	BR04
Use case name	Reserve a Vehicle and Confirm a Booking
Goal in context	Secure the customer's selected vehicle, complete an authenticated payment, and issue a confirmed booking without double allocation.
Scope	RideLanka FleetHire - Booking & Reservation Management
Level	User goal
Summary	A Registered Customer holds a vehicle for a selected rental period, supplies booking details, pays through the payment gateway, and receives a confirmed booking and digital receipt.
Priority	Critical - core customer and revenue flow
Primary actor	Registered Customer
Supporting actors	Payment Gateway; Email/SMS Service
Stakeholders and interests	Customer needs one reliable reservation; branch staff need accurate vehicle inventory; Finance needs a traceable payment; Operations needs only available vehicles booked.
Trigger	The Registered Customer selects a vehicle and rental period and chooses to reserve it.
Preconditions	The customer is signed in; the vehicle is AVAILABLE for the requested branch and dates; pickup and return dates are valid; required customer and contact data can be supplied.
Success postcondition	One confirmed booking owns the selected vehicle for the requested period, payment is SUCCESS, a digital receipt exists, and a booking-confirmation message is queued.
Failure guarantee	No vehicle is confirmed for two overlapping bookings; no receipt is issued for an unconfirmed payment; booking and payment outcomes remain traceable.
Related use cases	BR01 Search Available Vehicles; BR02 Check Vehicle Availability; BR03 Apply Promotion Code; PM01 Process Payment; FM02 Reserve Vehicle for Maintenance Window.
Implementation alignment	Core hold, booking, payment callback, confirmation, and receipt flows match the executable baseline. Guest checkout is a proposed feature and is not yet connected end to end.

03A Main Success Scenario

BR04 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The customer opens current availability for a vehicle category, branch, and date range.
2	The customer chooses a vehicle and requested add-ons and requests a hold.
3	The system releases expired holds, rechecks the selected vehicle's availability, and calculates the fare.
4	The system creates one time-limited, repeat-safe hold for the available vehicle.
5	The customer supplies the required driver and contact details.
6	The system validates and securely stores the details for the customer-owned hold.
7	The system creates the booking in PENDING_PAYMENT and presents the booking and fare summary.
8	The customer continues to the payment gateway and confirms the payment.
9	The payment gateway returns the payment result for the correct order.
10	The system verifies the provider, signature, order, amount, currency, result, and replay identity.
11	For an on-time successful payment, the system confirms the booking and the held vehicle, and records payment as SUCCESS.
12	The system issues a digital receipt and makes it available to the customer.
13	The system queues the approved booking-confirmation message and displays the confirmed booking.

03B Alternative and Exception Flows

BR04 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
3a	The selected vehicle is no longer available	The hold is not created; the system refreshes availability so the customer can choose again.
2a	Rental-period rules are invalid	The system explains the permitted minimum-rental and date-range rules.
7a	The hold expires before booking creation	The system rejects the stale hold and requires a new availability check.
10a	Signature, order, amount, or currency is invalid	The system rejects the callback; no booking or receipt is confirmed.
10b	The same callback is received again	The system returns the existing result without creating another booking, charge, or receipt.
11a	Successful payment arrives after the hold expires	The booking becomes PAYMENT_REVIEW, payment becomes LATE_SUCCESS, and no receipt is issued.
11b	Provider reports failure or cancellation	Payment becomes FAILED or CANCELLED and the booking remains unconfirmed.
12a	Confirmation or receipt storage fails	The confirmation operation is cancelled so a partial paid-booking result is not reported as complete.
13a	Message delivery is temporarily unavailable	The confirmed booking remains valid and the delivery attempt is available for controlled retry.

Business Rules

- A vehicle-date combination may have only one active confirmed booking.

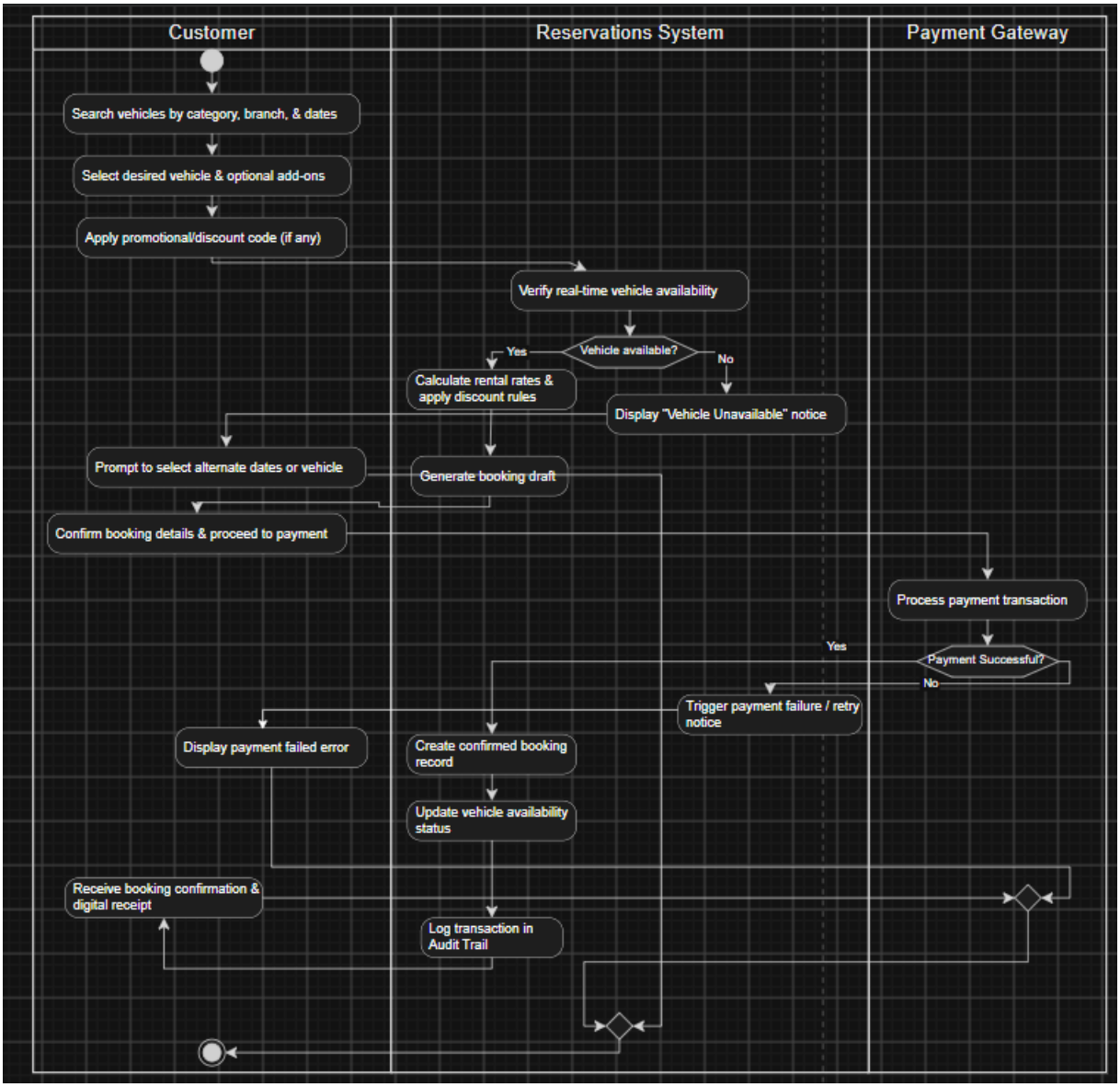
- Only AVAILABLE vehicles are bookable for the requested branch and dates.
- RideLanka FleetHire calculates the fare; a client-supplied total is not authoritative.
- A digital receipt is issued only after the linked payment result is accepted and the booking is confirmed.
- A customer may retrieve only bookings and receipts they own.

Special Requirements and Open Decisions

Concurrency: one winner for a same-vehicle race. Security and privacy: protect customer details, booking ownership, payment evidence, and receipt references. Accessibility: provide a usable list view as well as a visual availability calendar. Open issues: final hold duration, guest booking, change/cancellation limits, and refund rules.

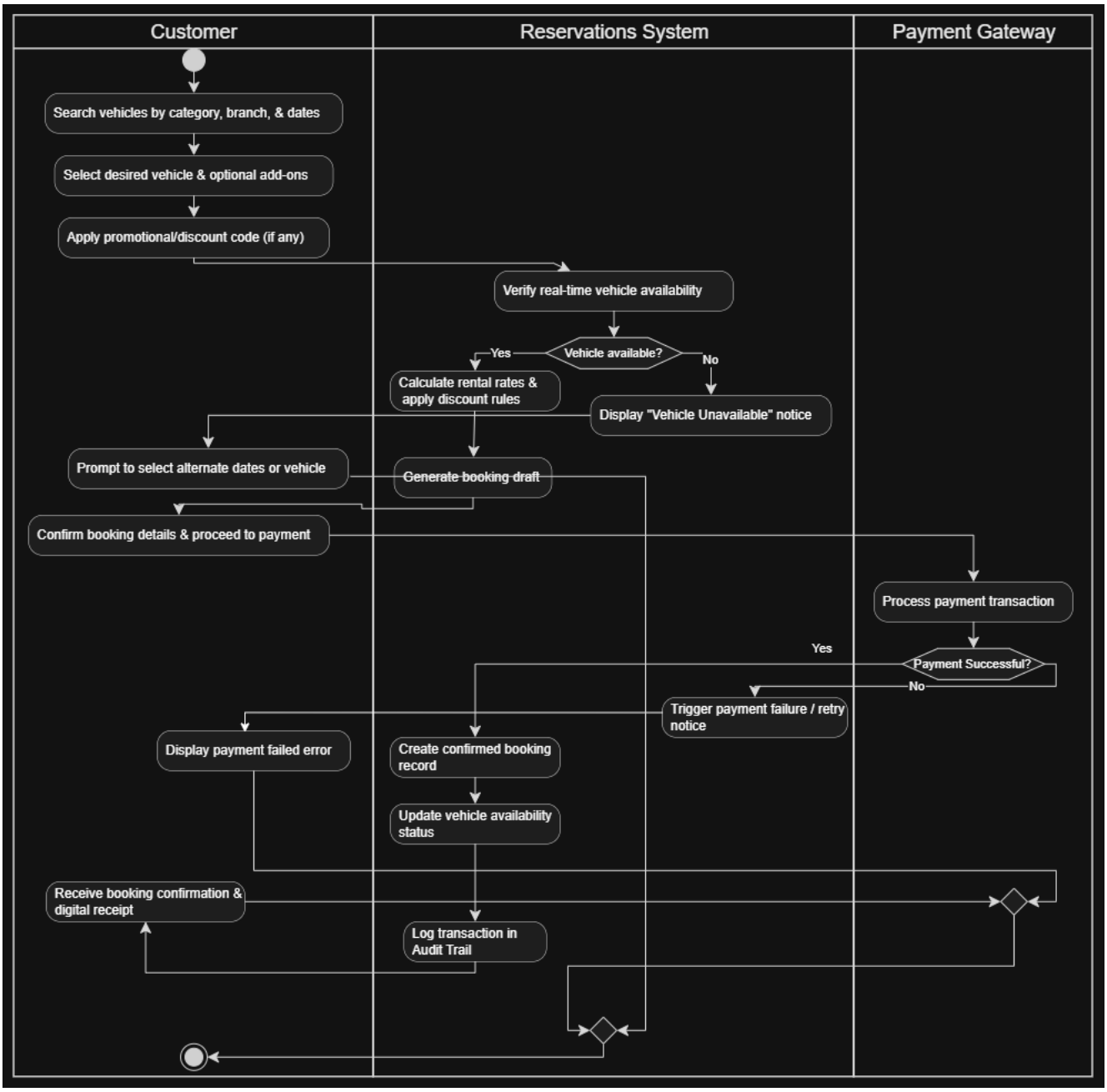
03C Draw.io Proof

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03D BR04 Activity Diagram - Clean Export

Swimlanes: Registered Customer and RideLanka FleetHire System, Payment Gateway; Email/SMS Service.



04 Payment management Activity Diagram

PM03 - Verify Payment Result Before Booking Confirmation

OWNERSHIP AND TRACEABILITY R M M D Bhagyamathi - IT25103733 | Core ownership: Payment management | Diagram use case: PM03

Use Case Definition

Use case ID	PM03
Use case name	Verify Payment Result Before Booking Confirmation
Goal in context	Accept a genuine payment-gateway result exactly once and prevent booking confirmation for an invalid, mismatched, failed, or late payment.
Scope	RideLanka FleetHire - Payment Management
Level	Subfunction supporting the booking goal
Summary	The payment gateway submits a callback; RideLanka FleetHire validates it, applies the correct payment and booking outcome, and retains accepted evidence.
Priority	Critical - financial integrity and booking control
Primary actor	Payment Gateway
Supporting actors	Finance Personnel (only when later review or reconciliation is required)
Stakeholders and interests	Customer needs the correct outcome; Finance needs auditable evidence; Booking needs a safe confirmation decision; the business must prevent duplicate revenue events and receipts.
Trigger	The payment gateway posts a payment callback for an existing RideLanka FleetHire checkout.
Preconditions	The payment and linked booking exist; the payment is normally PENDING; gateway credentials are configured; required callback fields are present.
Success postcondition	A valid on-time success is applied once, payment becomes SUCCESS, the eligible booking and vehicle are confirmed, a digital receipt is issued, and accepted callback evidence is retained.
Failure guarantee	An invalid or ineligible result does not confirm a booking or falsely mark it paid; its safe outcome is returned or routed for review.
Related use cases	PM01 Process Payment; PM02 Create Payment Checkout; PM04 Review or Reconcile Exception; PM05 Approve and Execute Refund; BR04 Reserve a Vehicle and Confirm a Booking.
Implementation alignment	Uses the current payment states and gateway callback fields. It intentionally does not claim a provider timestamp check or an automatically created finance case because neither exists in the current baseline.

04A Main Success Scenario

PM03 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The payment gateway sends merchant ID, order ID, payment ID, amount, currency, status code, and signature.
2	The system validates the required field formats, configured merchant, and callback signature.
3	The system maps the provider status to the corresponding RideLanka FleetHire outcome.
4	The system locates the payment by the provider order ID and prevents simultaneous processing of the same payment.
5	The system checks whether the callback represents a result already applied.
6	The system verifies the expected order, amount, currency, and linked booking.
7	For a successful result, the system checks that the linked vehicle hold is still valid.
8	The system confirms the booking and vehicle, issues a digital receipt, and records payment as SUCCESS.
9	The system retains the accepted callback evidence with the completed outcome.
10	The system returns a safe acknowledgement without exposing protected payment data.

04B Alternative and Exception Flows

PM03 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
2a	Merchant, format, or signature is invalid	The system rejects the callback before changing any payment, booking, or receipt state.
3a	Provider reports failed or cancelled	The system records FAILED or CANCELLED and issues no receipt.
3b	Provider reports chargeback after success	The system records CHARGEBACK for authorised operational and finance follow-up.
5a	The callback is a duplicate replay	The system returns the existing outcome without repeating any charge, booking, or receipt action.
6a	Order, amount, or currency does not match	The system rejects the callback; the current baseline does not invent an automatic finance case.
7a	Successful result arrives after hold expiry	Payment becomes LATE_SUCCESS, booking becomes PAYMENT_REVIEW, and no receipt is issued.
8a	The booking or receipt outcome cannot complete	The operation is cancelled so a partial successful result is not reported as final.

Business Rules

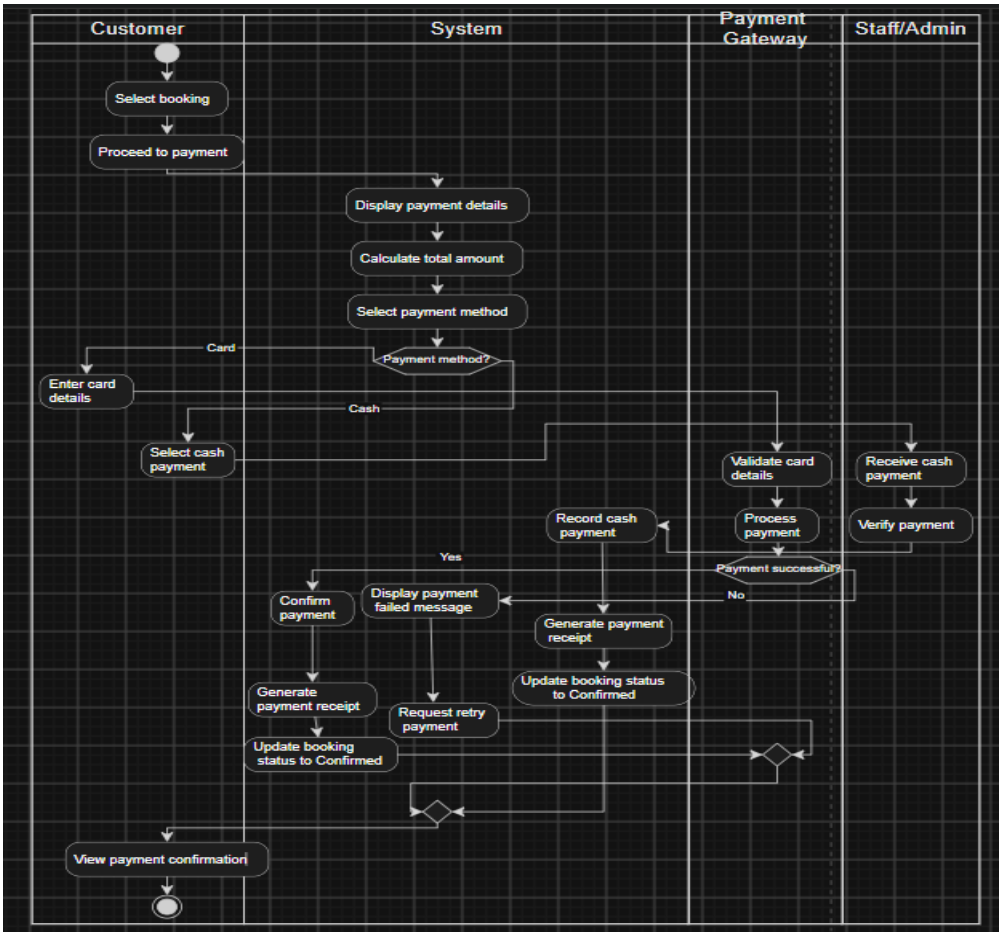
- The external payment gateway is the selected payment boundary for the current design.
- No digital receipt may be issued before an accepted successful result is linked to an eligible booking.
- The same provider result must not create duplicate business effects.
- Payment SUCCESS, FAILED, CANCELLED, LATE_SUCCESS, and CHARGEBACK remain distinct outcomes.
- Refund approval and execution are separate authorised Finance actions.

Special Requirements and Open Decisions

Security: protect credentials, verify signatures safely, and redact logs and responses. Reliability: repeated callbacks return one consistent business outcome. Auditability: accepted provider evidence is retained without later rewriting. Open issues: production gateway credentials, the live refund adapter, final LATE_SUCCESS disposition, and refund limits.

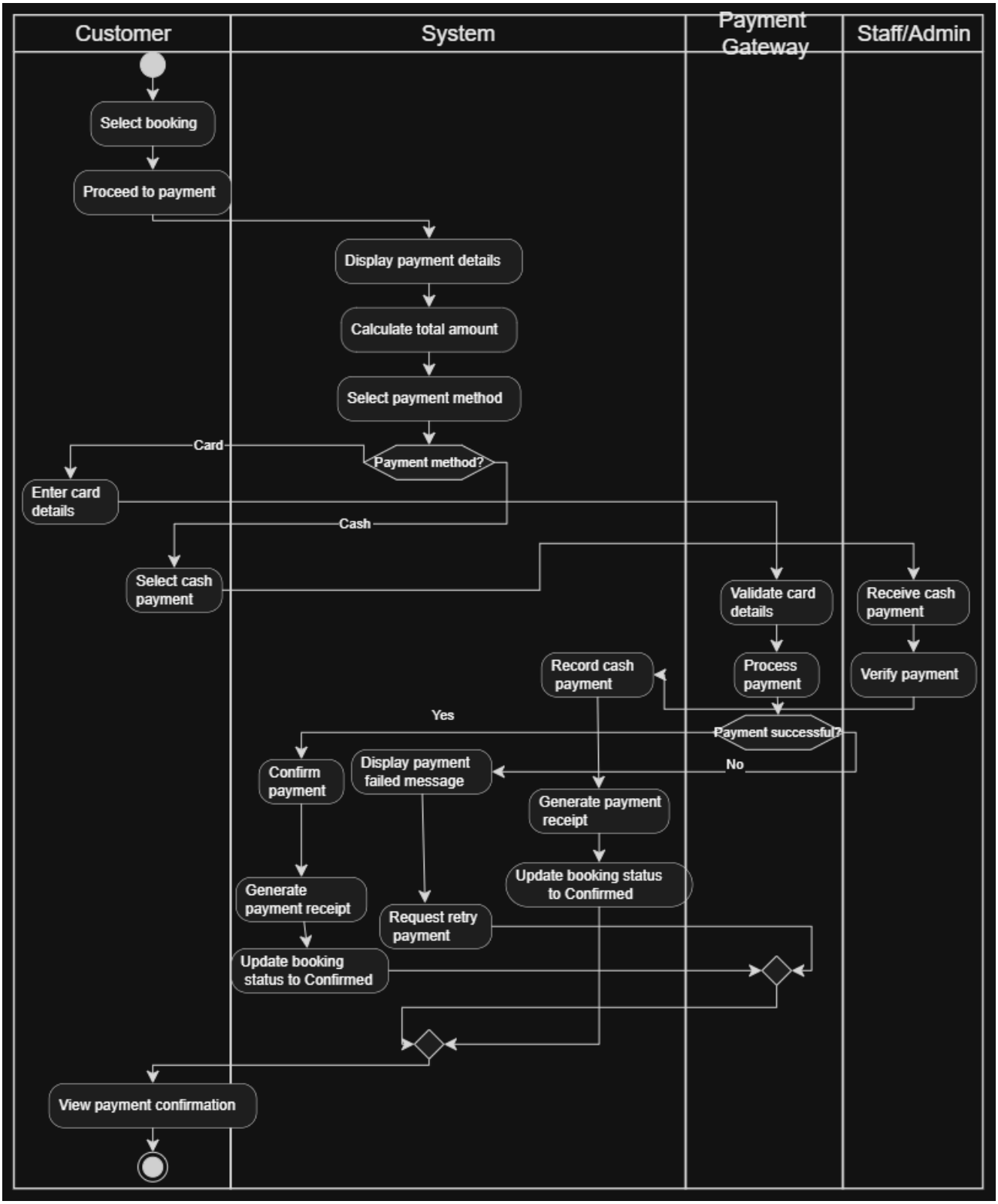
04C Draw.io Proof

EVIDENCE PURPOSE This space is reserved for a screenshot verifying that the activity diagram was authored and reviewed through Draw.io. Take a screenshot of your Draw.io canvas for this use case and insert it below.



04D PM03 Activity Diagram - Clean Export

Swimlanes: Payment Gateway and RideLanka FleetHire System, Finance Personnel (only when later review or reconciliation is required).



05 Feedback & support management Activity Diagram

FS02 - Submit and Assign a Customer Complaint Ticket

OWNERSHIP AND TRACEABILITY R M R D Rajapaksha - IT25103734 | Core ownership: Feedback & support management | Diagram use case: FS02

Use Case Definition

Use case ID	FS02
Use case name	Submit and Assign a Customer Complaint Ticket
Goal in context	Capture a customer complaint accurately and route it to the correct staff member for timely resolution.
Scope	RideLanka FleetHire - Feedback & Support Management
Level	User goal
Summary	A Registered Customer submits a complaint linked to a past rental; the system creates a ticket, and Customer Service staff assign it for resolution.
Priority	High - customer trust and satisfaction
Primary actor	Registered Customer
Supporting actors	Customer Service Staff
Stakeholders and interests	Customer needs their issue heard and tracked; Customer Service needs an organised ticket queue; management needs satisfaction and recurring-issue reporting.
Trigger	A customer experiences an issue with a rental and opens a complaint through their booking history.
Preconditions	The customer is signed in; the referenced booking belongs to the customer; a category and description can be supplied.
Success postcondition	A ticket is created in OPEN state, linked to the booking and customer, and either auto-routed or manually assigned to a Customer Service staff member.
Failure guarantee	No ticket is created without a valid customer-owned booking reference; no ticket is left unassigned beyond the routing step.
Related use cases	FS01 Submit Feedback and Rating; FS03 Update Ticket Status and Resolution; FS04 Generate Satisfaction Report; BR04 Reserve a Vehicle and Confirm a Booking.
Implementation alignment	Aligned to the current ticket creation and assignment baseline. Automated SLA-breach escalation is planned but not yet implemented.

05A Main Success Scenario

FS02 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The customer opens a past or current booking from their booking history.
2	The customer selects Raise a Complaint and chooses a category and description.
3	The system validates that the booking belongs to the signed-in customer and that required fields are present.
4	The system creates the ticket in OPEN state and links it to the booking and customer.
5	The system determines the appropriate queue based on category and branch.
6	The system assigns the ticket to an available Customer Service staff member or leaves it in the shared queue.
7	The system notifies the customer that the ticket was received with a reference number.
8	The system records the creation and assignment in the ticket's activity log.

05B Alternative and Exception Flows

FS02 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
2a	Category or description is missing	The system identifies the required field and does not create the ticket.
3a	Booking does not belong to the customer	The system denies ticket creation without revealing other customers' booking data.
3b	Booking cannot be found	The system reports that the booking cannot be located and creates no ticket.
6a	No staff member is currently available	The ticket remains OPEN in the shared queue for the next available staff member.
6b	Two staff members attempt to claim the same ticket	The system assigns the ticket to only one claimant and notifies the other of the conflict.
7a	Notification delivery is temporarily unavailable	The ticket remains valid and the delivery attempt is available for controlled retry.

Business Rules

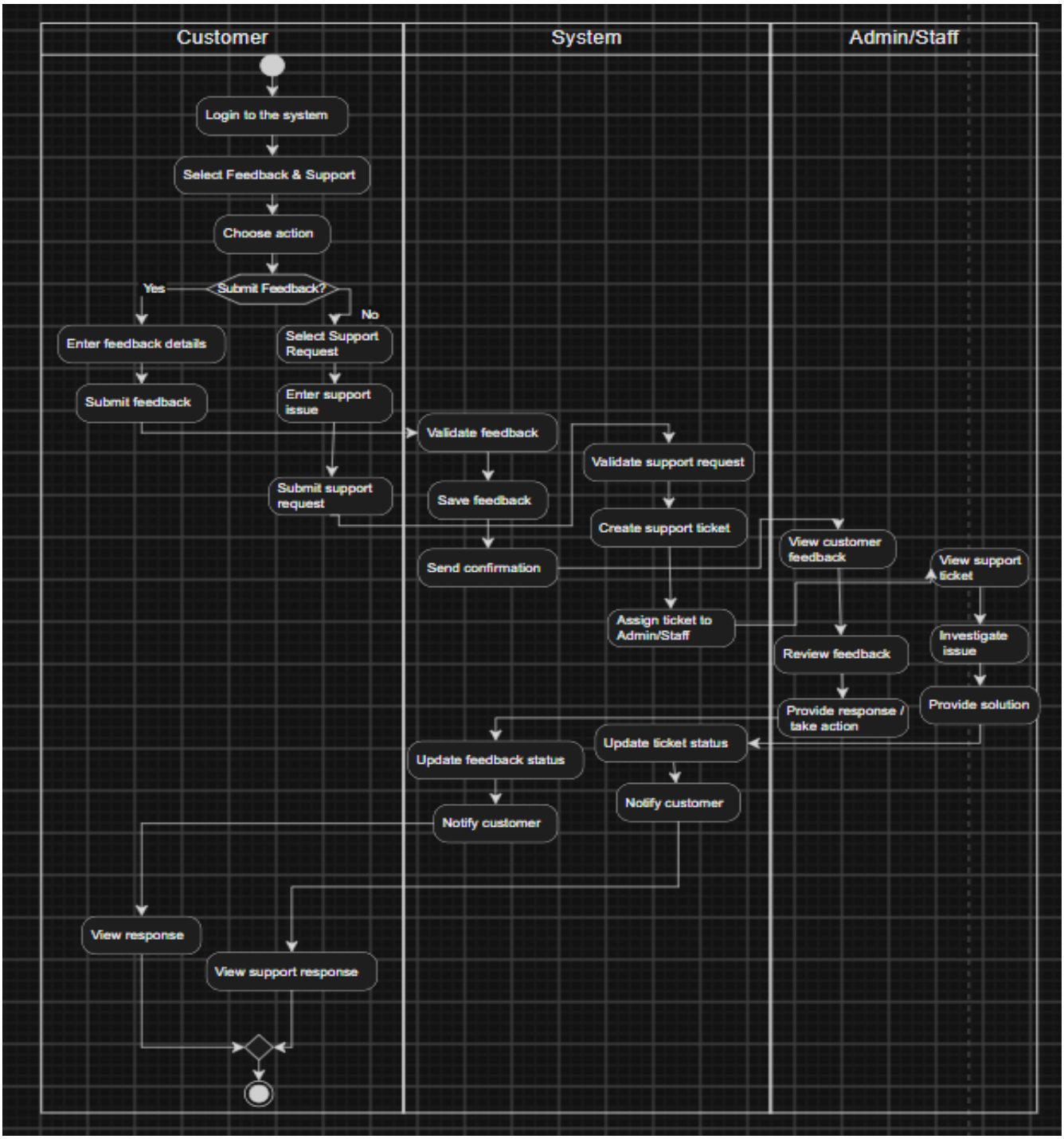
- A complaint ticket must reference a booking owned by the submitting customer.
- A ticket always starts in OPEN state and is never silently dropped.
- Only Customer Service staff and management may reassign or close a ticket.
- Every status change and assignment is recorded in the ticket's activity log.
- A customer may view only their own tickets and their current resolution status.

Special Requirements and Open Decisions

Usability: give the customer a clear reference number and expected next step. Privacy: staff see only the data needed to resolve the assigned ticket. Reliability: ticket creation and initial routing complete as one outcome. Open issues: SLA timers, escalation rules, and recurring-issue clustering remain future work.

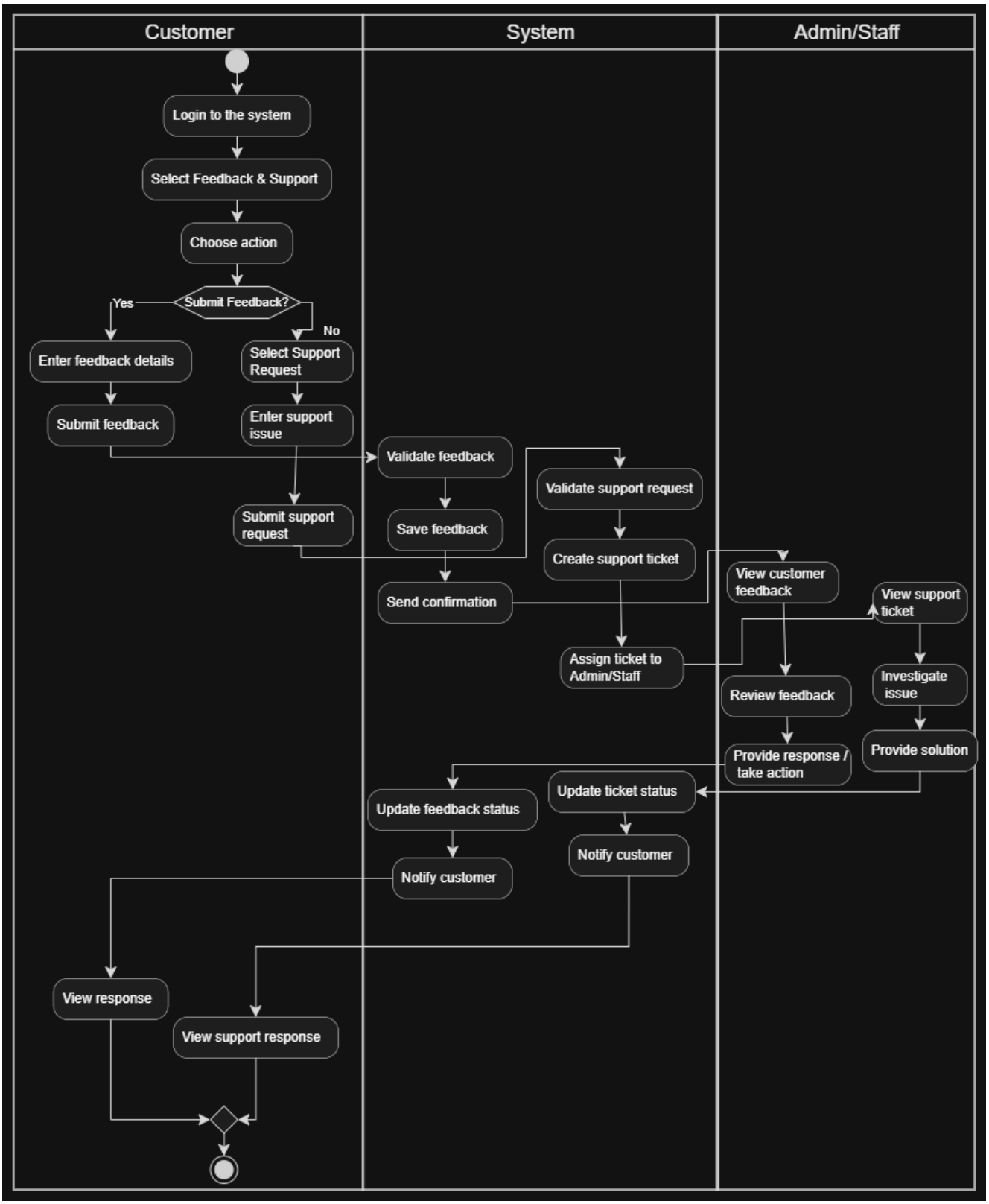
05C Draw.io Proof

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05D FS02 Activity Diagram - Clean Export

Swimlanes: Registered Customer and RideLanka FleetHire System, Customer Service Staff.



06 Fleet management Activity Diagram

FM03 - Schedule and Complete Vehicle Maintenance

OWNERSHIP AND TRACEABILITY K A K Dananjaya - IT25103735 | Core ownership: Fleet management | Diagram use case: FM03

Use Case Definition

Use case ID	FM03
Use case name	Schedule and Complete Vehicle Maintenance
Goal in context	Plan a vehicle's maintenance window, temporarily remove it from booking eligibility, and record the completed service history.
Scope	RideLanka FleetHire - Fleet Management
Level	User goal
Summary	A Maintenance Coordinator schedules a service window for a vehicle, the vehicle is blocked from new bookings during that window, and the coordinator records completion.
Priority	High - vehicle safety and fleet reliability
Primary actor	Maintenance Coordinator
Supporting actors	Fleet Management Staff
Stakeholders and interests	Maintenance Coordinator needs a reliable service plan; Reservations staff need accurate availability during the window; customers need to avoid being booked into an unserviced vehicle.
Trigger	The Maintenance Coordinator determines that a vehicle is due for scheduled servicing or repair.
Preconditions	The coordinator is signed in and active; the vehicle exists; the requested maintenance window does not conflict with a confirmed booking without prior reassignment.
Success postcondition	The maintenance record is created, the vehicle is excluded from new bookings for the window, and on completion the service history and vehicle status are updated.
Failure guarantee	No maintenance window is scheduled over an unresolved confirmed booking conflict; no partial service record is left without a final status.
Related use cases	FM01 View Fleet Maintenance Calendar; FM02 Reserve Vehicle for Maintenance Window; VM03 Mark a Vehicle Out of Service; VM04 View Vehicle Status History.
Implementation alignment	Aligned to the current maintenance-scheduling baseline. Automatic customer reassignment during a conflicting window is a proposed enhancement and is not yet implemented.

06A Main Success Scenario

FM03 authoritative actor-system interaction sequence

Step	Actor / System Interaction
1	The Maintenance Coordinator opens the selected vehicle's maintenance calendar.
2	The system displays current bookings, existing maintenance windows, and vehicle status.
3	The coordinator selects a maintenance window, service type, and notes.
4	The system checks the window against confirmed bookings for the vehicle.
5	The system creates the maintenance record in SCHEDULED state and blocks new bookings for the window.
6	On arrival at the vehicle, the coordinator marks the maintenance as IN_PROGRESS.
7	Upon completion, the coordinator records the service performed, parts used, and outcome.
8	The system updates the maintenance record to COMPLETED and appends it to the vehicle's service history.
9	The system restores the vehicle's availability for booking after the window ends.

06B Alternative and Exception Flows

FM03 guarded branches, safe recovery, and system responses

The following guarded outcomes preserve the last valid business state, direct recoverable issues to the appropriate retry point, and prevent any partial or unaudited result.

Ref	Condition	System Response
3a	Service type or notes are missing	The system identifies the required field and does not create the record.
4a	The window conflicts with a confirmed booking	The system reports the conflict and requires the coordinator to resolve it before scheduling.
5a	Vehicle is already OUT_OF_SERVICE	The system allows scheduling but does not change the existing out-of-service status.
6a	Coordinator attempts to start maintenance early	The system requires the scheduled window to have begun before allowing IN_PROGRESS status.
7a	Completion details are incomplete	The system keeps the record IN_PROGRESS and requests the missing details.
8a	The completion or history record cannot be saved	The operation is cancelled and the prior maintenance state remains authoritative.

Business Rules

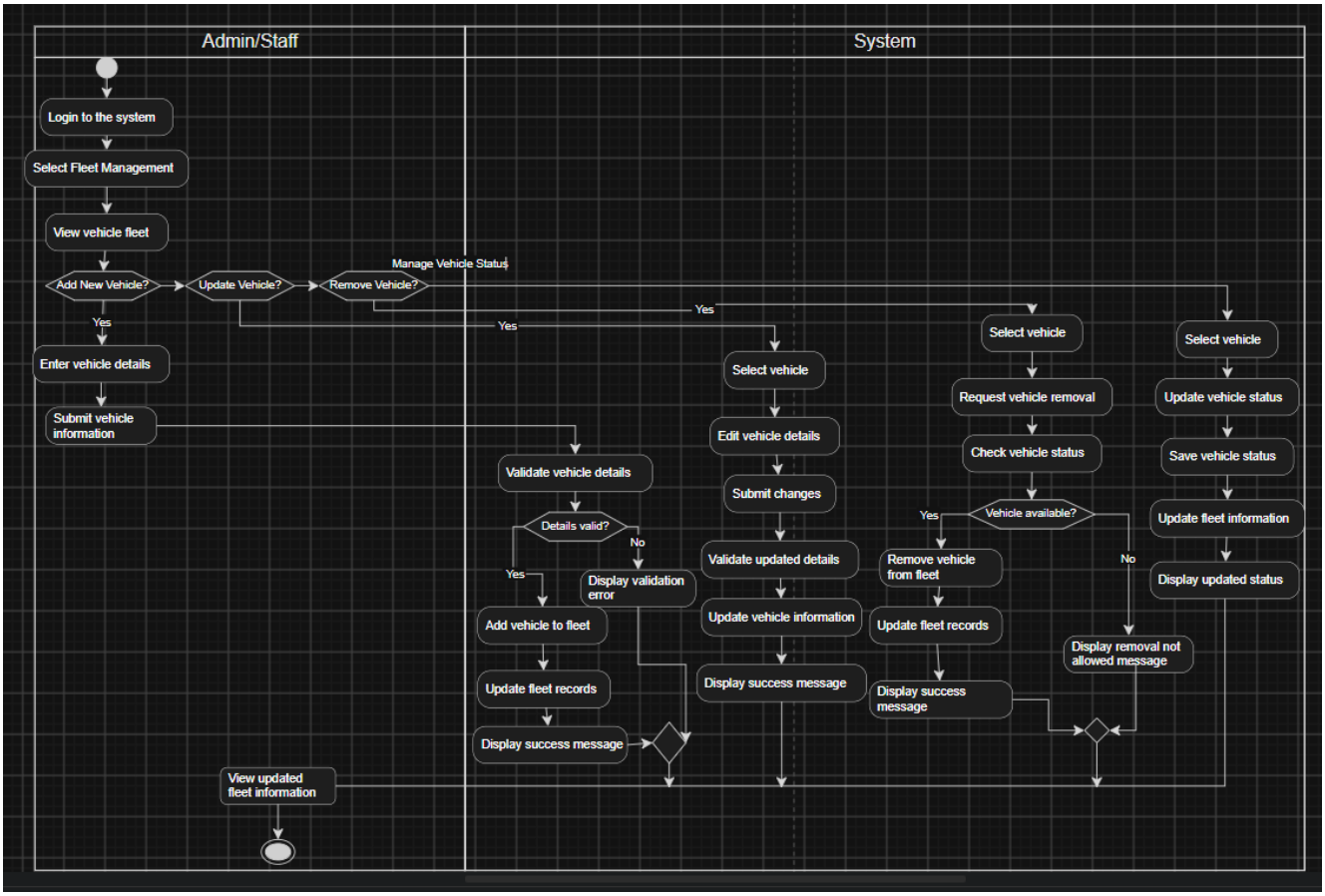
- A vehicle cannot be booked for dates that overlap an active maintenance window.
- Only Maintenance Coordinators and Fleet Management staff may create or update maintenance records.
- A maintenance record must reach COMPLETED or CANCELLED; it cannot be left indefinitely IN_PROGRESS.
- Completed service history is retained for the lifetime of the vehicle record.
- Availability is restored automatically only after a maintenance window ends and no OUT_OF_SERVICE status remains.

Special Requirements and Open Decisions

Reliability: status update and service-history append complete as one outcome. Usability: conflict messages should identify the overlapping booking. Concurrency: two coordinators cannot schedule overlapping windows for the same vehicle. Open issues: automated customer reassignment, parts-inventory linkage, and recurring maintenance schedules remain future work.

06C Draw.io Proof

EVIDENCE PURPOSE This space is reserved for a screenshot verifying that the activity diagram was authored and reviewed through Draw.io. Take a screenshot of your Draw.io canvas for this use case and insert it below.



06D FM03 Activity Diagram - Clean Export

Swimlanes: Maintenance Coordinator and RideLanka FleetHire System, Fleet Management Staff.

